



Jaya Bharathi, MDS,
Shweta Mittal, MDS,
Sanjay Tewari, MDS,
Shikha Tewari, MDS,
Jigyasa Duhan, MDS,
Pankaj Sangwan, MDS, and
Vinay Kumar, MDS

CONSORT RANDOMIZED CLINICAL TRIAL

Effect of the Piezoelectric Device on Intraoperative Hemorrhage Control and Quality of Life after Endodontic Microsurgery: A Randomized Clinical Study

SIGNIFICANCE

Piezosurgery resulted in better hemorrhage control during surgery and quality of life of patients in the first week postsurgery. Piezoelectric bone surgery might be a predictable alternative to conventional surgical carbide burs and curettes to enhance treatment acceptance by the patients because of its improved postoperative sequelae.

From the Post Graduate Institute of Dental Sciences, Pt.B.D. Sharma University of Health Sciences Rohtak, Haryana, India

Address requests for reprints to Dr Shweta Mittal, Department of Conservative Dentistry and Endodontics, Postgraduate Institute of Dental Sciences, Rohtak, Haryana, India 124001.
E-mail address: shwetagoelendo@gmail.com
0099-2399/\$ - see front matter

Copyright © 2021 American Association of Endodontists.
<https://doi.org/10.1016/j.joen.2021.04.013>

The postoperative quality of life of the patient depends on the level of pain, tissue swelling, drug intake, chewing ability, phonetics, and conduction of normal daily routine activities¹. Pain and swelling are the common sequelae observed immediately in the postsurgical period. The assessment of the quality of life of patients after the postsurgical period is a valuable tool in predicting postoperative symptoms². Only a few prospective studies (1 longitudinal and 3 randomized controlled trials) have evaluated the quality of life of patients after endodontic surgery²⁻⁵.

Piezoelectric bone surgery is an innovative technology permitting selective cutting of hard tissue while sparing soft tissues. Piezoelectric surgery works similar to a dental scaler with 3 to 6 times higher power. It has many advantages like precise cutting, preservation of neurovascular structures, minimal

thermal damage to bone, greater visibility, hemorrhage control, cavitation effect, and an overall improvement in healing⁶.

Patients who had osteotomies with the piezoelectric device showed significantly less postoperative edema, trismus, and pain as well as analgesic intake compared with conventional rotary instruments in mandibular third molar surgery⁷. *In vivo* animal studies have found better initial neo-osteogenesis with piezosurgery compared with conventional burs^{8–10}. Piezosurgery was found to be better than conventional curettage in the enucleation of cysts with no cyst recurrence^{11,12}.

Also, 1 of the major concerns during periapical surgery is the attainment of adequate hemostasis. This will improve visibility, reduce operating time, facilitate surgical procedures (root-end resection, root end preparation, and filling), and minimize surgical hemorrhage. The bleeding inside the bony crypt also must be controlled for proper sealing of retrofilling material, which may affect the outcome of endodontic surgery. Few studies in oral surgery also reported that piezosurgery results in less bleeding compared with standard surgical burs^{13,14}.

The application of the piezoelectric device has been used recently in 2 case reports to create a bony window on buccal bone in cases of the intact buccal cortical plate with large periapical lesions^{15,16}. The management of both hard and soft tissues with the piezoelectric device might limit postoperative symptoms and discomfort, which may enhance the patient's quality of life and acceptance of endodontic surgery, although hemorrhage control during surgery may enable the surgeon to perform the surgical procedures with greater ease. Hence, the aim of this study was to evaluate whether the use of piezoelectric surgery might have an impact on quality of life during the first week postsurgery. The secondary aim was to compare intraoperative hemorrhage control between piezosurgery versus a carbide bur.

MATERIALS AND METHODS

The clinical trial was performed after approval by the institutional ethics committee (PGIDS/IEC/2018/26) in the postgraduate section of the Department of Conservative Dentistry and Endodontics, Postgraduate Institute of Dental Sciences, Rohtak, India. Patients requiring endodontic surgery were enrolled from December 2018–June 2019. The trial was registered on clinicaltrials.gov with ID: NCT04358887.

Patient Inclusion

The eligibility criteria included patients who were medically fit for oral surgical procedures (American Society of Anesthesiologists 1 or 2), patients 16–55 years of age, and patients with the diagnosis of symptomatic/asymptomatic apical periodontitis in previously treated teeth with a large (≥ 5 mm) periapical lesion and with radiographic evidence of an intact buccal cortical plate on cone-beam computed tomographic (CBCT) imaging. The root canal treatment of the involved teeth was adequate as observed on both periapical radiography and CBCT imaging. Maxillary and mandibular anterior teeth with periradicular lesions of endodontic origin and a size < 20 mm were included. The tooth had an appropriate final restoration with no coronal leakage.

Patients with systemic diseases (diabetes mellitus, uncontrolled hypertension grade III, hepatic/renal disease, or systemic bleeding disorders), pregnant and lactating women, and patients on anticoagulant/antiplatelet drugs were excluded. Teeth that were unrestorable, fractured/perforated teeth, teeth with apicomarginal defects, and teeth with deep pockets (probing depth > 4 mm) were also excluded. Information regarding the surgical procedure and its associated risks and benefits and the necessary follow-up care were provided to all the patients in written form, and those who gave consent were included in the study.

Sample Size

Sample size calculation was based on postoperative pain (which was considered the most important manifestation affecting the quality of life)². Based on data obtained from a previous study with a power of 80%, a significance level equal to 5%, a clinically significant difference of 0.8, and pooled standard deviation of 0.824, 16 patients for each group were needed². Assuming an attrition rate of 20%, a total of 40 patients (21 women and 19 men with 74 teeth) were included, 20 patients in each group.

Randomization

Forty patients (74 teeth) were randomized equally to both the piezo ($n = 20$) and the control groups ($n = 20$) using a block of an equal number of opaque envelopes containing concealed codes. Patients were asked to pick the envelope before the start of the surgical procedure. Patients were blinded for surgical allotment. The sham surgical setup was created by the placement of a piezoelectric surgical device for both groups during surgery

for blinding of independent observers for intraoperative bleeding assessment. Operator blinding was not possible because of the use of a piezoelectric surgical device in 1 group.

Surgical Procedure

All clinical procedures were performed by a single operator. Except for incision, flap elevation, and suturing, all surgical procedures were performed under an operating microscope (OPMI PICO; Carl Zeiss, Göttingen, Germany; with $f_{\text{tube}} = 170$, $f_{\text{lens}} = 250$, and $m_{\text{eyepiece}} = 12.5$) in an operating room. Low magnification of 0.4–0.6 $\times$ was used for orientation and the alignment of surgical tips, midrange magnification of 1.0–1.6 $\times$ for surgical procedures and retropreparations, and high magnification of 1.6–2.5 $\times$ for observation of fine details like inspection of the resected root surface.

A full-thickness mucoperiosteal flap was reflected under sterile conditions after administering local anesthesia with lidocaine 2% with epinephrine 1:1,00,000. After flap elevation, in the piezo group, bone cutting was performed using the piezoelectric surgical device (Woodpecker Ultrasurgery USII, Guangxi, China) with a power input of 100–240 V (50 Hz/60 Hz), a working frequency of 24–36 kHz, and an irrigation flow rate of 25–110 mL/min. The entry point on intact bone was estimated preoperatively using CBCT imaging. The osteotomy was performed using a US6 piezoelectric tip (Woodpecker Ultrasurgery USII, Guangxi, China) set in a bone mode in a linear to-and-fro motion with continuous ejection of sterile saline preventing heat buildup (Fig. 1A). After that, refinement of the ragged edges was performed using a UL 4 diamond insert. Piezo tips with different angles like UL1 (120°), UL2, and UL5 (95°) (Woodpecker Ultrasurgery USII, Guangxi, China) were used to separate the lining of the cyst or granulation tissue from the bone cavity (Fig. 1B).

In the control group, after flap elevation, bone cutting was performed using a surgical carbide bur (#557 SL; Brasseler USA, Savannah, GA) (Fig. 1D). Complete debridement of the bony lesion was performed by surgical curettes (Fig. 1E).

After curettage, the principal surgeon and 2 blinded observers recorded the level of hemorrhage independently. Once the recording was completed for additional hemorrhage control during surgery, epinephrine-soaked cotton pellets were applied topically as and when required.

Root-end resection of 2–3 mm with a 0°–10° angle was performed using a US6

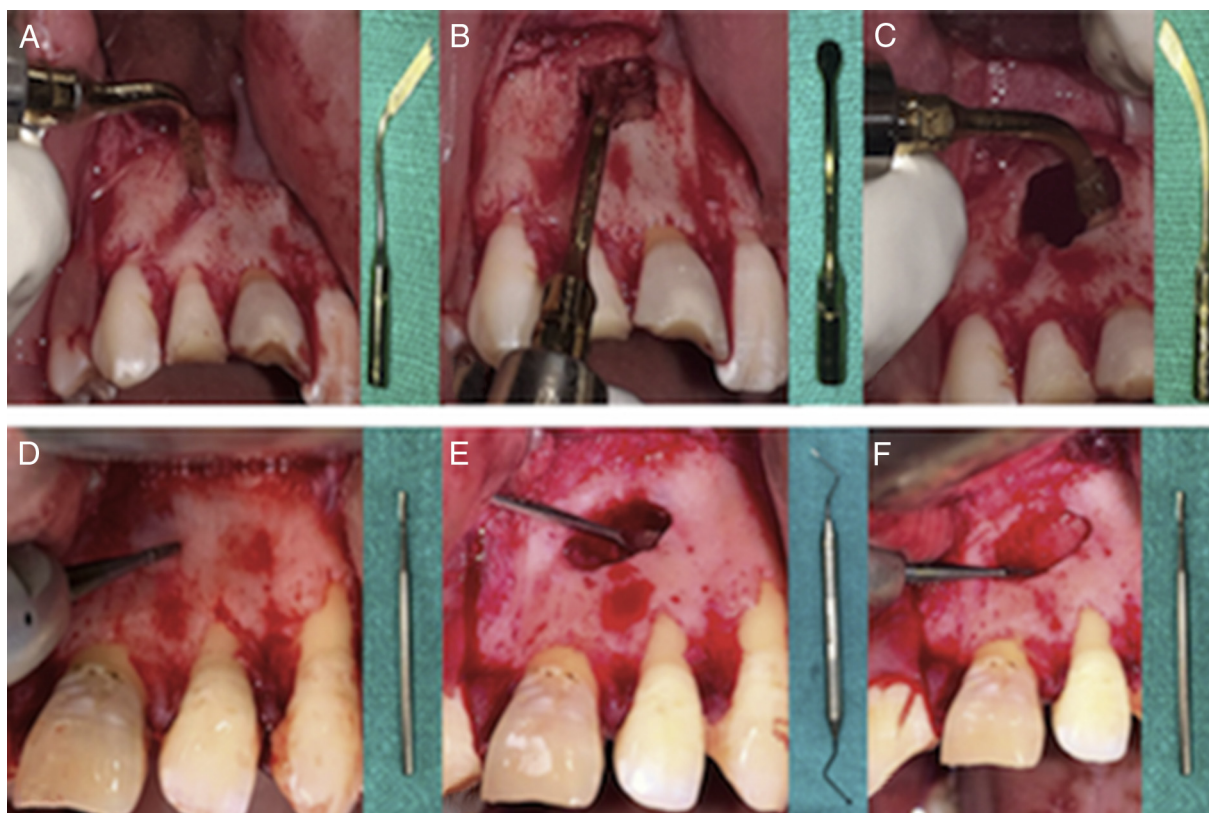


FIGURE 1 – (A–C) Intraoperative pictures of the piezo group bone cutting using (A) a US 6 piezoelectric insert, (B) granulation tissue removal using a UL 1 insert, and (C) root-end refinement using a UL 4 insert. (D–F) Intraoperative pictures of control group bone cutting using a (D) surgical carbide bur, (E) granulation tissue removal using surgical curettes, and (F) root-end resection using a carbide bur.

piezo insert (Woodpecker Ultrasurgery USII, Guangxi, China) and refined with a UL4 diamond insert (Woodpecker Ultrasurgery USII, Guangxi, China) in the piezo group, whereas a cylindrical surgical carbide bur was used in the control group (Fig. 1C and F).

Root-end preparation of approximately 3 mm was made along the long axis of the root with a piezoelectric ultrasonic unit (P5 Booster, Suprasson Neutron; Acteon Inc, Mt Laurel, NJ) with double-angled S12-7D retro tips (Satelec, Sylmar, CA) coated with diamond abrasives in both the groups. The root end was stained

with 2% methylene blue to evaluate any missed anatomic irregularities like isthmuses, fins, and so on. Root-end filling was performed with mineral trioxide aggregate (Dentsply, York, PA). The wound site was closed and sutured with a 5-0 silk suture after irrigating the defect properly.

Postoperative Instructions

The patients were advised not to gargle and to avoid hard, hot, and spicy foods; heavy physical activity; smoking; and tooth brushing on the day of surgery. Patients were

prescribed chlorhexidine digluconate 0.2% mouthwash to rinse their mouth twice daily up to 1 week postsurgically. Nonsteroidal analgesics (ibuprofen 400 mg) were advised for the relief of pain/swelling control after the surgical procedure if needed. No antibiotics were prescribed. Patients were recalled after 5 days for suture removal.

Evaluation Parameters

Hemorrhage control during surgery was assessed by the surgeon and 2 evaluators who were blinded to the groups and assigned



FIGURE 2 – Intraoperative pictures showing Vy et al scoring. (A) 0 = no hemorrhage control, (B) 1 = intermittent hemorrhage control, and (C) 2 = complete hemorrhage control.

scores of 0 (no hemorrhage control, continuous or intermittent bleeding that obscured the surgical site to perform root-end filling procedures), 1 (slight but apparent intermittent bleeding that allowed to perform root-end filling procedures), or 2 (complete hemorrhage control that allowed root-end filling procedures)¹⁷ (Fig. 2A–C). Scores 0 and 1 were graded as inadequate and 2 as adequate hemorrhage control. The assessment of blinded observers was only taken into account while recording the bleeding control, and in case of any difference, the consensus was obtained by discussion.

The quality of life of patients was evaluated for limitations of oral and general functions, pain, and other symptoms. For limitations of oral and general functions, a modified version of the patient's perception questionnaire provided by Del Fabbro et al was used⁴. Current instruments of quality of life mainly focus to assess the impact on the quality of life of various chronic oral conditions and treatments. The patient's perception questionnaire focuses on postoperative signs and symptoms after surgical procedures that may influence the patient's quality of life⁴. The questionnaire was given to all patients to evaluate postoperative restraint in function (mouth opening, chewing, speaking, sleeping, daily routine, and work) and the existence of any symptoms (swelling, nausea, and bad taste/breath), which were evaluated by a 5-point Likert type scale ranging from 1–5 (1 = not at all, 2 = very little, 3 = some, 4 = quite a bit, and 5 = very much). For pain assessment, a visual analog scale (VAS) was used where 0 = no pain and 100 = the worst possible pain. All the patients were given training regarding filling out the questionnaire and VAS and recording symptoms. The patients were given a validated questionnaire in his or her local language, and the recording of grading of symptoms was explained to each participant by the single analyzer. Furthermore, they were advised to contact the analyzer in case of any doubts or clarification. They were also given the telephone number of the analyzer in case of doubt while filling out the form at home, and patients were called at least twice during a 1-week period as a reminder and also to enquire whether they were facing any doubts. Finally, patients were asked to note the number of analgesics taken each day until 7 days. Patients were asked to fill out the questionnaire every day, starting from the day of surgery up to 7 days postoperatively. Questionnaires were returned by the patient after 7 days.

TABLE 1 - The Descriptive Statistics of the Patients Included in the Study

Table 1: The percentage of patients being emergent during the first postoperative week. Data indicate a significant difference between the 2 groups in respect to age.

REFERENCES

1. Gutmann J, Harrison JW. *Surgical Endodontics*. 1st ed. Boston, MA: Blackwell Scientific Publications; 1991.
2. Tsesis I, Shoshani Y, Givol N, et al. Comparison of quality of life after surgical endodontic treatment using two techniques: a prospective study. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 2005;99:367–71.
3. Del Fabbro M, Taschieri S, Weinstein R. Quality of life after microscopic periradicular surgery using two different incision techniques: a randomized clinical study. *Int Endod J* 2009;42:360–7.
4. Del Fabbro M, Ceresoli V, Lolato A, Taschieri S. Effect of platelet concentrate on quality of life after periradicular surgery: a randomized clinical study. *J Endod* 2012;38:733–9.
5. Meschi N, Fieuws S, Vanhoenacker A, et al. Root-end surgery with leucocyte- and platelet-rich fibrin and an occlusive membrane: a randomized controlled clinical trial on patients' quality of life. *Clin Oral Investig* 2018;22:2401–11.
6. Hennes P. Piezoelectric bone surgery: a review of the literature and potential applications in veterinary oromaxillofacial surgery. *Front Vet Sci* 2015;2:8.
7. Al-Moraissi EA, Elmansi YA, Al-Sharaee YA, et al. Does the piezoelectric surgical technique produce fewer postoperative sequelae after lower third molar surgery than conventional rotary instruments? A systematic review and meta analysis. *Int J Oral Maxillofac Surg* 2016;45:383–91.
8. Preti G, Martinasso G, Peirone B, et al. Cytokines and growth factors involved in the osseointegration of oral titanium implants positioned using piezoelectric bone surgery versus a drill technique: a pilot study in minipigs. *J Periodontol* 2007;78:716–22.
9. Yang BE, Girod S. Efficacy of bone healing in calvarial defects using piezoelectric surgical instruments. *J Craniofac Surg* 2014;25:149–53.
10. Blaskovic M, Gabrić D, Coleman NJ, et al. Bone healing following different types of osteotomy: scanning electron microscopy (SEM) and three-dimensional SEM analyses. *Microsc Microanal* 2016;22:1170–8.
11. Yaman Z, Suer BT. Clinical comparison of ultrasonic surgery and conventional surgical techniques for enucleating jaw cysts. *Int J Oral Maxillofac Surg* 2013;42:1462–8.
12. Kocyigit ID, Atil F, Alp YE, et al. Piezosurgery versus conventional surgery in radicular cyst enucleation. *J Craniofac Surg* 2012;23:1805–8.
13. Spinelli G, Lazzeri D, Conti M, et al. Comparison of piezosurgery and traditional saw in bimaxillary orthognathic surgery. *J Craniomaxillofac Surg* 2014;42:1211–20.
14. Landes CA, Stübinger S, Rieger J, et al. Critical evaluation of piezoelectric osteotomy in orthognathic surgery: operative technique, blood loss, time requirement, nerve and vessel integrity. *J Oral Maxillofac Surg* 2008;66:657–74.
15. Hirsch V, Kohli MR, Kim S. Apicoectomy of maxillary anterior teeth through a piezoelectric bony-window osteotomy: two case reports introducing a new technique to preserve cortical bone. *Restor Dent Endod* 2016;41:310–5.
16. Lee SM, Yu YH, Wang Y, et al. The application of "bone window" technique in endodontic microsurgery. *J Endod* 2020;46:872–80.
17. Saunders WP. A prospective clinical study of periradicular surgery using mineral trioxide aggregate as a root-end filling. *J Endod* 2008;34:660–5.

18. Kim S. Modern endodontic practice: instruments and techniques. *Dent Clin North Am* 2004;48:1–9.
19. Pandurić DG, Bago I, Katanec D, et al. Comparison of Er:YAG laser and surgical drill for osteotomy in oral surgery: an experimental study. *J Oral Maxillofac Surg* 2012;70:2515–21.
20. Barone CM, Jimenez DF, Yule GJ, Strauch B. Analysis of bone formation after cranial osteotomies with a high-speed drill. *J Craniofac Surg* 1997;8:466–70.
21. Kerawala CJ, Martin IC, Allan W, Williams ED. The effects of operator technique and bur design on temperature during osseous preparation for osteosynthesis self-tapping screws. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 1999;88:145–50.
22. Slade GD, Spencer AJ. Development and evaluation of the oral health impact profile. *Community Dent Health* 1994;11:3–11.
23. Haefeli M, Elfering A. Pain assessment. *Eur Spine J* 2006;15:S17–24.
24. Patil C, Jadhav A, Rajanikanth K, et al. Piezosurgery vs bur in impacted mandibular third molar surgery: evaluation of postoperative sequelae. *J Oral Biol Craniofac Res* 2019;9:259–62.
25. Chang HH, Lee MS, Hsu YC, et al. Comparison of clinical parameters and environmental noise levels between regular surgery and piezosurgery for extraction of impacted third molars. *J Formos Med Assoc* 2015;114:929–35.
26. Amaya F, Izumi Y, Matsuda M, Sasaki M. Tissue injury and related mediators of pain exacerbation. *Curr Neuropharmacol* 2013;11:592–7.
27. Watt DG, Horgan PG, McMillan DC. Routine clinical markers of the magnitude of the systemic inflammatory response after elective operation: a systematic review. *Surgery* 2015;157:362–80.
28. Gopal I, Morkel J, Titinchi F. Comparison of a piezoelectric and a standard surgical handpiece in third molar surgery. *Oral Surg* 2019;12:30–4.
29. Vickers FJ, Baumgartner JC, Marshall G. Hemostatic efficacy and cardiovascular effects of agents used during endodontic surgery. *J Endod* 2002;28:322–3.
30. Vy CH, Baumgartner JC, Marshall JG. Cardiovascular effects and efficacy of a hemostatic agent in periradicular surgery. *J Endod* 2004;30:379–83.
31. Scarano A, Artese L, Piattelli A, et al. Hemostasis control in endodontic surgery: a comparative study of calcium sulfate versus gauzes and versus ferric sulfate. *J Endod* 2012;38:20–3.
32. Menéndez-Nieto I, Cervera-Ballester J, Maestre-Ferrín L, et al. Hemostatic agents in periapical surgery: a randomized study of gauze impregnated in epinephrine versus aluminum chloride. *J Endod* 2016;42:1583–7.
33. Peñarrocha-Oltra D, Menéndez-Nieto I, Cervera-Ballester J, et al. Aluminum chloride versus electrocauterization in periapical surgery: a randomized controlled trial. *J Endod* 2019;45:89–93.
34. Peñarrocha-Oltra D, Soto-Peñaloza D, Peñarrocha-Diogo M, et al. Hemostatic agents in endodontic surgery: a randomized controlled pilot study of polytetrafluoroethylene strips as an adjunct to epinephrine impregnated gauze versus aluminum chloride. *J Endod* 2019;45:970–6.
35. Kimura Y, WilderSmith P, Matsumoto K. Lasers in endodontics: a review. *Int Endod J* 2000;33:173–85.
36. Schütz S, Egger J, Kühl S, et al. Intraosseous temperature changes during the use of piezosurgical inserts in vitro. *Int J Oral Maxillofac Surg* 2012;41:1338–43.